

DVWA SQL Injection Assessment Report

Project

Web Application Security Assessment using DVWA & Burp Suite

Objective

Perform a manual SQL Injection assessment against DVWA using Burp Suite Repeater and validate vulnerable application behavior through parameter manipulation.

Environment

- Kali Linux
- DVWA
- Burp Suite Community Edition
- Firefox Browser

Assessment Steps

1. Configured Burp Suite Proxy.
2. Intercepted HTTP requests sent to the DVWA SQL Injection module.
3. Sent captured requests to Burp Repeater.
4. Tested multiple SQL Injection payloads.
5. Compared HTTP responses and application behavior.

Test Results

Baseline Request

Payload:

```
id=1
```

Observation:

- Returned a single valid user record.

Malformed Input

Payload:

```
id=1 '
```

Observation:

- Application returned HTTP 500 Internal Server Error.

- Indicates backend query failure due to unsanitized input.

Boolean-based SQL Injection

Payload:

```
id=1' OR '1'='1
```

Observation:

- Returned multiple user records.
- Successfully bypassed the intended query condition.

UNION-based SQL Injection

Payload:

```
id=1' UNION SELECT user,password FROM users#
```

Observation:

- Extracted usernames and password hashes from the database.
- Confirmed successful UNION-based SQL Injection.

Risk Assessment

The application is vulnerable to SQL Injection because user input is incorporated directly into SQL queries without proper parameterization.

Recommended Remediation

- Use parameterized queries (prepared statements)
- Implement server-side input validation
- Apply least-privilege database permissions
- Use centralized error handling
- Avoid exposing detailed database errors

Conclusion

The assessment successfully validated SQL Injection vulnerabilities using manual testing techniques in Burp Suite. The project demonstrates practical experience with HTTP interception, request manipulation, SQL Injection validation, and vulnerability documentation within a controlled web application security testing environment.